

Does a concrete layer harm established trees?

Testing a public claim about the Upper Barrakka Gardens, Valletta, against the scientific literature

THE CLAIM UNDER REVIEW

“Given its temporary nature, it will not cause damage to the tree.”

Ambjent Malta, to the Times of Malta, 1 October 2026, on concrete placed in tree planters.
Context: waterproofing planters to stop water entering the Lascaris War Rooms tunnels.

VERDICT: NOT SUBSTANTIATED

Plausible, but no evidence published

Version 1.1 · 2 October 2026 · revised after reading the full Times of Malta article

Status: draft for right of reply (Ambjent Malta, Fondazzjoni Wirt Artna)

Prepared from public sources. No site visit or tree assessment was carried out.

Repository: github.com/leandergrech/Mizien

TL;DR

Ambjent Malta told the Times of Malta that concrete placed in the tree planters at Upper Barrakka will not damage the trees **because it is temporary**: it stays only until a waterproofing membrane is installed. Ambjent also said prolonged cover would be a concern, and that concrete lying directly on the roots has since been removed. The works protect the historic Lascaris War Rooms from water ingress. That aim is legitimate and is not questioned here. **The question is whether the tree-safety statement is backed by evidence.**

- 1 The claim is conditional, and its logic points the right way.** Ambjent itself says prolonged cover would be a problem, for reasons (less water, bark and nutrient transport, decay) that match the research on sealed soils.
- 2 But “temporary” is undefined, and the claim covers only the concrete.** No duration is given. The article also reports that soil was removed from the planter and that concrete lay around the roots. Root exposure, contact with fresh concrete and how it was removed are not addressed.
- 3 No evidence has been published.** No species, tree ages, dates, arborist report or monitoring data. The “five planters, no tree died” statement is anecdote, and decline often takes 3–5+ years to show.
- 4 The science: mixed for permanent sealing, silent on temporary cover.** Studies of permanent sealing range from no harm in a five-year trial to clear harm in holm oak. We found *no* study of short-term concrete contact with roots.
- 5 Verdict: not substantiated.** Plausible for a short, well-managed cover, but not shown for these trees. Ambjent responded quickly and was candid about the risk.

THIS CLAIM



SUPPORTED

LARGELY
SUPPORTED

NOT
SUBSTANTIATED

MISLEADING

CONTRADICTED

Undefined

How long “temporary” lasts: no duration published

3–5+ yrs

Typical delay before construction damage shows in the crown

None found

Studies of short-term concrete contact with roots; published arborist data

28

Water-infiltration points the works are meant to fix

WHAT WOULD MOVE THE VERDICT UP

A stated, short duration with dates; roots shown to have been kept covered and moist and not cut; clean removal of the concrete; arborist inspection before and after; a membrane design that drains; monitoring of crown health for several years.

WHAT WOULD MOVE IT DOWN

Concrete in contact with roots or bark for days or weeks; roots left exposed in hot, dry weather or cut; bark damaged on removal; sealed planters that waterlog; monitoring data withheld or never collected.

IN THIS REPORT

- 1 The claim and what we could verify
- 2 Method and how to read this report
- 3 How concrete could harm a tree
- 4 What the studies found

- 5 Where the science disagrees
- 6 Testing the claim at Upper Barrakka
- 7 Verdict and requests for evidence
- 8 Limitations

1

The claim and what we could verify

The claim comes from a Times of Malta article by Nicole Meilak (1 October 2026), read in full. Its headline says the concrete layer will not damage the trees. **Ambjent Malta’s own words are narrower:** the concrete is temporary, so it will not cause damage, and prolonged cover would be a concern. We treat Ambjent’s wording as the claim and the headline as the paper’s summary of it. Statements from Fondazzjoni Wirt Artna (FWA), which runs the tunnels, are labelled as FWA’s.

Source	What it says	Our access	Status
Times of Malta, 1 Oct 2026 [3]	Concrete laid at the bottom of tree planters; soil removed from at least one planter. Ambjent: the concrete is temporary, remaining only until a waterproofing membrane is installed, so it will not damage the tree; harm could arise if left prolonged. Concrete directly on the roots has since been removed (photo caption). No duration given.	Read in full (supplied PDF).	The claim itself
Same article, quoting FWA	Severe flooding of the war rooms from storm and irrigation water leaking from the gardens. Ambjent made every effort to save existing trees, especially old ones. Every leak point addressed.	Read in full.	Partner body’s statement
Newsbook, 1 Oct 2026, reporting FWA [1]	Waterproofing includes “tanking” of planters. 28 water-infiltration points addressed. Five planters treated so far with no tree dying or removed.	Search excerpt only; page blocks automated access.	Partner body’s statement
MaltaToday, c. Dec 2025, quoting FWA’s executive chairman [2]	Waterproofing membrane under the concrete; mulch on top; large pots of flowers and plants; trunks and crowns untouched.	Read in full.	Earlier phase of works

FAIRNESS NOTE

The seepage problem is real and serious: storm and irrigation water entered a complex whose parts date to 1570, 1640 and 1715, and FWA says some areas are permanently lost. Sealing planters is a recognised way to stop leaks. Ambjent responded within days, stated the tree risk plainly and had concrete removed from the roots. This is a trade-off between two goods. The issue examined here is narrower: **whether the statement that the trees will not be harmed is supported by evidence.**

2

Method and how to read this report

Question. Does pouring concrete over or near the roots of an established tree harm it, and does the research support Ambjent Malta’s position that a *temporary* layer will not?

Evidence. Targeted searches of peer-reviewed literature and credible grey literature (university extension services, park services, arboricultural reviews), run in October 2026. We favoured studies on *established* trees with a control. Few studies address poured concrete specifically, so most evidence comes from pavements and soil sealing generally, and we found none on short-term, temporary cover.

Evidence grades used in the tables:

A	Controlled or randomised experiment on trees	B	Observational field study with a control or gradient
C	Review or professional guidance	D	Assertion or anecdote with no data

Verdicts follow a five-point scale defined in Appendix A. Differing scientific views are shown side by side in Section 5 rather than averaged away. Where the science has not settled, we say so instead of choosing a side.

3 How concrete could harm a tree

Concrete can affect an established tree in three largely separate ways. The literature treats them very differently, and it is easy to conflate them.

- 1. During the works.** Digging, regrading and compaction cut or crush roots, and any root left exposed can dry out quickly, especially in hot or windy weather. Guidance is to avoid hot, dry conditions and to cover exposed roots with soil, mulch or damp burlap as soon as possible [13, 17]. Park-service and extension guidance names compaction and root severance as the commonest causes of construction-related decline, with symptoms sometimes taking years to appear [12, 13].
- 2. After the works: soil sealing.** A sealed surface reduces rain and irrigation reaching the soil and slows gas exchange with the air. Sealed soils can run warmer; roots can be killed above roughly 40°C [11]. Reported consequences include lower soil oxygen, higher CO₂ and fewer fine roots [4, 5].
- 3. At the trunk and in the chemistry.** Fresh cement-based material leaches strongly alkaline water for a time [14, 15]. Industry guidance also cautions against burying the root flare [18]. We found no controlled study of concrete placed against roots or trunks, temporary or permanent. **Duration is the unknown:** the sealing studies measure months to years.

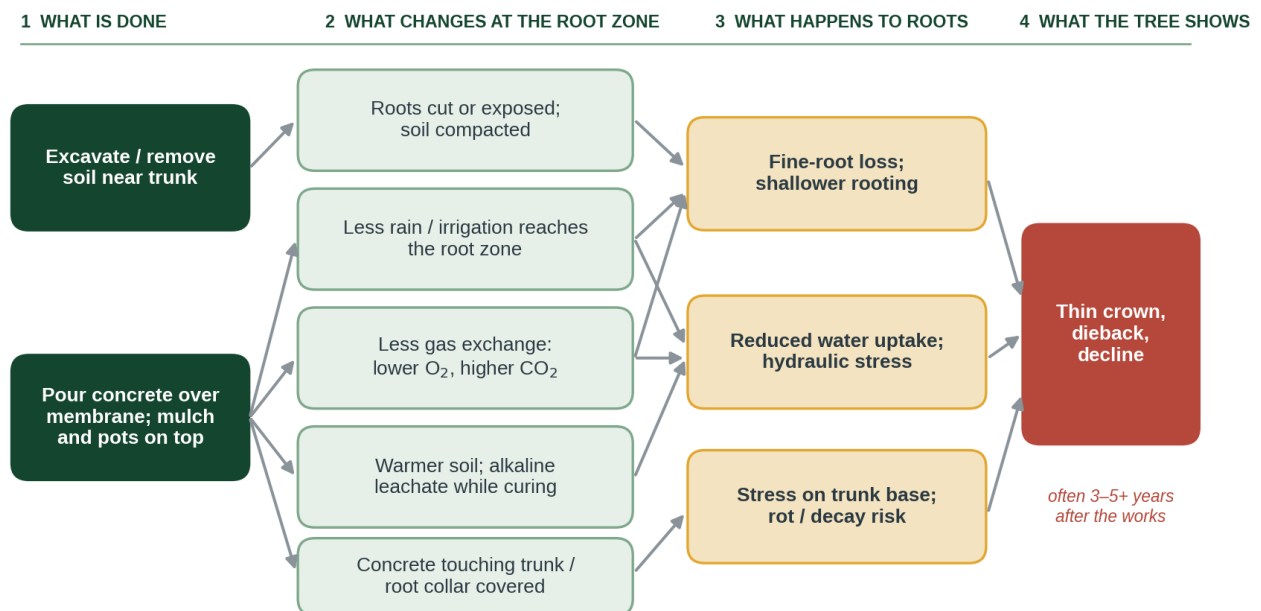


Figure 1. Pathways by which concrete works could harm an established tree. Which pathways operate, and how strongly, depends on species, age, the fraction of the root zone sealed, distance from the trunk, soil and water supply.

WHY OXYGEN MATTERS

Roots generally perform best above about 10% soil oxygen. Estimated minimum thresholds are roughly 3% for root survival and 5–10% for growth [16]. Soil under impervious concrete had lower oxygen than under pervious concrete or bare soil, especially when wet [4].

4

What the studies found

Figure 2 places each source on a qualitative scale from “adverse effect reported” to “no adverse effect detected”. **Placement is our judgement of each source’s headline finding. It is not a statistical meta-analysis.** Diamonds mark sources we know only through another paper’s summary.

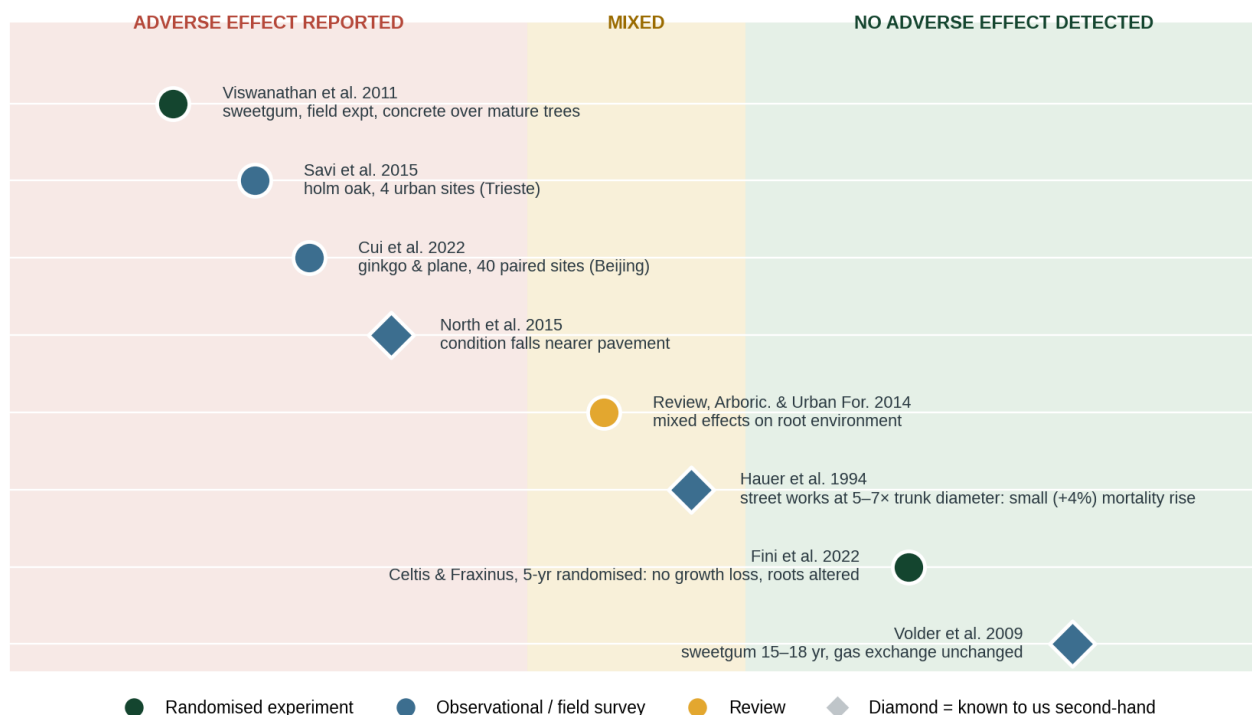


Figure 2. Evidence map. Fini et al. found root morphology altered by pavements but no drought stress or above-ground growth penalty; Hauer et al. found only a small mortality rise for works at distance.

Source	Design and trees	Key finding	Reading
Viswanathan et al. 2011 [4] A	Field experiment: concrete installed over pre-existing mature sweetgum on clay.	Lower soil oxygen under impervious concrete (esp. wet), higher CO ₂ ; far less fine-root length than unpaved controls. Pervious concrete did not help on low-permeability clay.	Adverse (roots)
Fini et al. 2022 [5] A	5-year randomised trial, 48 trees in 1 m ² pits (planted 2012); <i>Celtis australis</i> , <i>Fraxinus ornus</i> ; wetter temperate site.	No evidence of drought stress or above-ground growth loss. Impermeable surfaces cut soil moisture, especially winter–spring. Root morphology altered, likely via CO ₂ .	No above-ground effect; roots altered
Savi et al. 2015 [6] B	Four urban sites with different impervious cover; holm oak (<i>Quercus ilex</i>), Trieste, Italy.	More impervious cover: greater drought stress, reduced leaf gas exchange, more xylem cavitation.	Adverse (Mediterranean species)
Cui et al. 2022 [7] B	40 paired paved/vegetated sites in Beijing; <i>Ginkgo biloba</i> , <i>Platanus orientalis</i> .	Growth increments lower on paved land: ginkgo -44.5% diameter, -31.9% height; plane -31.7% and -60.1%.	Adverse
Volder et al. 2009 [8] ♦ B	Field study, 15–18-year-old sweetgum (as reported in [5]).	Leaf gas exchange, fluorescence and water relations did not differ between paved and unpaved.	No effect detected
Hauer et al. 1994 [10] ♦ B	Street trees near construction (as summarised in [9]).	Works at 5–7 trunk diameters from the tree: about +4% mortality, -5% condition rating.	Small effect at distance

Source	Design and trees	Key finding	Reading
North et al. 2015 ♦ B	Cited in a TreeNet summary [11].	Tree health and condition fall as the pavement edge nears the trunk.	Adverse with proximity
Arboric. & Urban For. review 2014 [9] C	Review of root-management strategies.	Pavement effects on the root environment are mixed. Soil moisture can be higher under pavement. Benefits of pervious paving not consistently shown.	Mixed
NPS / extension guidance [12, 13] C	Professional guidance, no new data.	Construction damage may take 3–5+ years to show. Rule of thumb: ~20% root loss tolerated, ~40% likely fatal.	Context
Wirt Artna via Newsbook [1] D	Five treated planters; species, ages and dates not stated.	No tree has died or been removed.	Anecdote

Grades: A controlled experiment · B observational · C review/guidance · D anecdote. ♦ known to us second-hand.

Reading across the evidence

- **No study we found tests the Upper Barrakka configuration:** concrete over a waterproofing membrane around older trees in a Mediterranean climate, and none tests a *temporary* cover.
- **The closest evidence points the wrong way.** Holm oak, the Mediterranean species most studied, showed greater hydraulic stress where impervious cover was high. That evidence is observational.
- **The best-controlled evidence is more reassuring but transfers poorly.** It concerns relatively young trees in small pits at a temperate site averaging about 1,100 mm of rain a year, far wetter than the Maltese Islands.
- **Roots react more consistently than crowns, and crowns lag.** Several studies find altered roots without an immediate canopy penalty, which is why short monitoring windows can mislead.

5 Where the science disagrees

Six questions drive the dispute. For each we set out the best evidence on each side, why the findings may differ, and what that means for Upper Barrakka.

Q1 Does sealing cause drought stress in established trees?

UNRESOLVED

EVIDENCE OF HARM

Holm oak among more impervious cover showed greater drought stress, reduced gas exchange and more xylem cavitation [6]. Ginkgo and oriental plane on paved ground grew markedly less than neighbours on vegetated ground [7].

EVIDENCE OF LITTLE OR NO HARM

A five-year randomised trial on established *Celtis* and *Fraxinus* found no drought stress or above-ground growth penalty [5]. Soil moisture can be higher under pavement because evaporation is reduced [9]. Gas exchange in 15–18-year sweetgum was unchanged [8].

WHY THEY MAY DIFFER, AND WHAT IT MEANS FOR UPPER BARRAKKA

Species differ; the Italian trial was in a wetter climate with small pits; the harm studies are observational and may be confounded. **For Barrakka:** the holm-oak data are the closest match to Malta's climate, but nobody has published tree-level data for these trees.

Q2 Does a temporary cover matter?

EVIDENCE GAP

EVIDENCE OF HARM

Exposed roots can dry out quickly in hot or windy weather [13, 17]. Fresh concrete leachate is strongly alkaline [14, 15]. Soil oxygen under impervious concrete was lower than under other surfaces, and roots need oxygen (about 3% is a survival minimum) [4, 16].

EVIDENCE OF LITTLE OR NO HARM

No direct evidence. Ambjent's premise is that harm comes from prolonged cover. The sealing studies measured effects over months to years [4, 5], which is consistent with duration mattering. *That is our inference, not a tested result.*

WHY THEY MAY DIFFER, AND WHAT IT MEANS FOR UPPER BARRAKKA

We found no study of short-term concrete contact with roots, so both sides here are reasoning, not data. **For Barrakka:** the outcome depends on how long, in what weather, and whether exposed roots were protected. None of this is reported.

Q3 Does it matter whether the surface is permeable?

LEANING: LITTLE CONSISTENT BENEFIT

EVIDENCE OF HARM

Impervious cover blocks infiltration and gas exchange, and permeable paving has been proposed as a mitigation [5, 9].

EVIDENCE OF LITTLE OR NO HARM

Benefits of pervious paving are not consistently shown [9]. Pervious concrete over low-permeability clay did not improve the root zone of mature trees [4]. Permeable options did not enhance above-ground growth [5].

WHY THEY MAY DIFFER, AND WHAT IT MEANS FOR UPPER BARRAKKA

Narrow pavements may let air and water diffuse in from the edges anyway [9]. **For Barrakka:** the design is impervious on purpose, because the membrane exists to stop water reaching the rock below. The real question is where the trees' water will come from.

Q4 Are mature trees more or less vulnerable than young ones?

UNRESOLVED

EVIDENCE OF HARM

Large established trees transpire more and may deplete soil water if infiltration is blocked. Their root responses are the least certain [5]. Reported dieback occurred in holm oak and linden older than 20 years [5].

EVIDENCE OF LITTLE OR NO HARM

Young trees often establish *faster* in paved pits, helped by reduced evaporation (e.g. plane trees grew larger in paved than unpaved soil) [5].

WHY THEY MAY DIFFER, AND WHAT IT MEANS FOR UPPER BARRAKKA

Most experiments use young trees. In-situ studies on old trees carry confounding. **For Barrakka:** Wirt Artna describes the trees as older and well established, the least-studied and most uncertain category.

Q5 Does concrete alkalinity harm roots?

PROBABLY MINOR;
UNDER-STUDIED

EVIDENCE OF HARM

Fresh cement-based material leaches very alkaline water: crushed non-carbonated concrete reaches pH 13–14 [14], and early-age concrete raised water pH to about 11 in the lab [15]. High pH can limit nutrient availability.

EVIDENCE OF LITTLE OR NO HARM

Soil buffers the effect and clay soils slow the alkaline front [14]. In the Italian trial pavements did not change soil chemical traits [5]. We found no field study showing alkaline-leachate death of an established tree.

WHY THEY MAY DIFFER, AND WHAT IT MEANS FOR UPPER BARRAKKA

Most evidence is lab-scale or from recycled aggregate. **For Barrakka:** fresh concrete touching roots, even briefly, is the least-studied scenario; the Maltese substrate is already calcareous (general knowledge, not reviewed here).

Q6 How much root loss can a tree tolerate?

HEURISTICS, NOT
THRESHOLDS

EVIDENCE OF HARM

Guidance suggests ~20% root loss is tolerated and ~40% probably fatal [13]. Severing roots over about 5 cm (2 in) can reduce stability and uptake [12].

EVIDENCE OF LITTLE OR NO HARM

Street works at 5–7 trunk diameters raised mortality only about 4% [10]. Tolerance depends on species, health and *which* roots are lost.

WHY THEY MAY DIFFER, AND WHAT IT MEANS FOR UPPER BARRAKKA

Rules of thumb are not validated thresholds. **For Barrakka:** excavation depth, distance to trunk and the share of the root zone under the slab are all unreported. The article reports soil removed from the planter.

6

Testing the claim at Upper Barrakka

We split the public statements into checkable sub-claims and rated each against the evidence above.

Sub-claim	Said by	What the evidence shows	Rating
A. “Given its temporary nature, it will not cause damage to the tree”	Ambjent Malta (Times of Malta)	Plausible in direction, since sealing harm builds with time. But “temporary” has no stated duration, no tree-specific assessment is given, and the statement does not address the soil removal or root exposure the article reports.	NOT SUBSTANTIATED
B. Prolonged cover would be a concern: water, bark, nutrient transport, decay	Ambjent Malta	Matches mechanisms in the literature (table below). Candid and accurate as far as it goes.	CONSISTENT WITH RESEARCH
C. Concrete directly on the roots has since been removed	Times of Malta / Ambjent	The photo and caption show concrete cleared from directly around the trunk, leaving a mound of loose soil and stone. Root and bark condition after removal is not reported.	PARTLY VERIFIED
D. Ambjent made every effort to save old trees	FWA (Times of Malta)	No method statement, root-protection plan or arborist report seen.	UNVERIFIED
E. Five treated planters, no tree has died or been removed	FWA (Newsbook)	Death is a late, coarse signal; decline may take 3–5+ years. No dates, species or condition data (Figure 3).	WEAK EVIDENCE
F. Earlier phase: membrane, mulch and pots; trunks and crowns untouched	FWA chairman (MaltaToday, c. Dec 2025)	The Oct 2026 article reports soil removed from at least one planter, so “untouched” can only refer to trunks and crowns. Root condition is not addressed.	UNVERIFIED / AMBIGUOUS

HEADLINE VERSUS STATEMENT

The headline states the point without condition. Ambjent Malta’s own words are conditional on the concrete being temporary and acknowledge a risk if it stays. This report assesses the conditional statement. A reader who sees only the headline gets a stronger reassurance than Ambjent actually gave.

Ambjent Malta’s reasoning, checked

Ambjent’s stated concern	What the research says	Match
Could restrict water availability around the trunk	Sealed surfaces reduce infiltration and soil moisture in several studies [4, 5]. Effect on established-tree drought stress is contested (Section 5, Q1).	CONSISTENT
Could affect the bark and nutrient transport	Fine-root loss reduces uptake [4]; guidance cautions against burying the root flare [18]. No controlled study of concrete against bark.	PLAUSIBLE
Could contribute to decay	Guidance links root and trunk-base stress to decay risk [12, 18]. No controlled study of concrete contact.	PLAUSIBLE
Only if left in place for a prolonged period	No published threshold for how long cover is tolerated. Delayed symptoms (3–5+ years) mean short-term outcomes can mislead [12].	UNTESTED

Why “no tree has died” is weak evidence

Guidance literature stresses that trees damaged in construction can look healthy for years before crown thinning and dieback appear [12]. An absence of deaths therefore says little unless the works are old enough, and the trees are monitored with something finer than survival: canopy density, leaf area, shoot growth, root-collar condition.

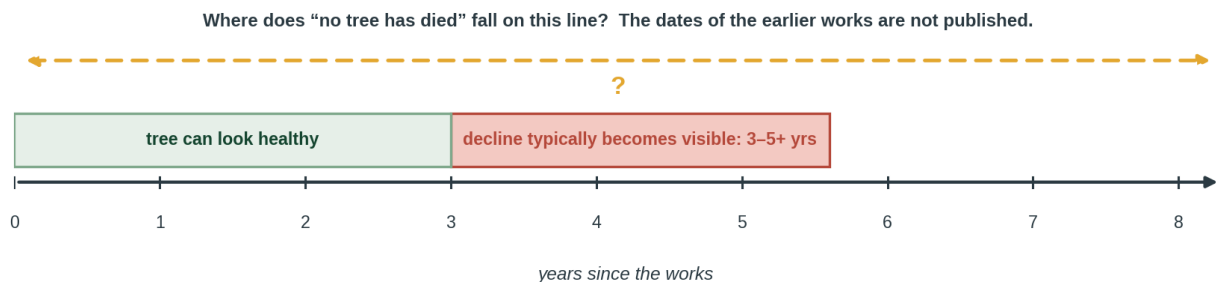


Figure 3. Schematic only. The decline window reflects guidance in [12, 13], not a measured curve for these trees.

WHAT THE PHOTOGRAPH CAN AND CANNOT SHOW

The article’s photograph (caption: the concrete directly under the trunk has been removed) shows a mound of loose soil and stone at the base, surrounded by a grey concrete layer across the planter floor, with the planter’s soil gone. It was taken after removal. It cannot show how long the concrete was in contact, the condition of the roots and bark, or whether roots were cut. Those are the quantities that determine the outcome.

7

Verdict and requests for evidence

VERDICT

NOT SUBSTANTIATED

Plausible, but not shown. Confidence: moderate. Limited by absent site data and the absence of studies on temporary cover.

Why. (1) The claim is conditional and its logic is sound in direction: the research suggests harm from sealing builds with time, and Ambjent itself names prolonged cover as the concern. (2) But no duration is given for “temporary”, and the statement is silent on the soil removal and root exposure the article reports, which guidance literature treats as a major risk. (3) No tree-specific evidence has been published. (4) For permanent sealing the literature is mixed, with holm-oak data pointing toward harm; for temporary cover we found no studies at all.

What this verdict does not say. It does not say the trees have been or will be damaged, that the works were unjustified, or that Ambjent acted in bad faith. Ambjent responded within days, stated the risk plainly and had concrete removed from the roots. A defensible statement would read: “*The concrete was in place for [N days]; exposed roots were protected by [method]; an arborist inspected on [date]; we will monitor [indicators] for [years].*” That turns a reassurance into something checkable.

Evidence we are asking for

- ☐ Species, estimated age and trunk diameter of every tree in a treated planter.
- ☐ For each planter: when the concrete was placed and when it was removed (duration of contact).
- ☐ How the soil was removed, whether roots were exposed, cut or damaged, and how exposed roots were protected (kept covered and moist, work in cooler weather).
- ☐ How the concrete was cleared from the roots and trunk base, and photographs of root and bark condition afterwards.
- ☐ The membrane design: drainage, the substrate that replaces the soil, and how the trees will be watered.

- ☐ Any pre- and post-works arborist report.
- ☐ Dates, species and monitoring records behind the “five planters, no tree died” statement.
- ☐ Whether any tree is protected under the Trees and Woodlands (Protection) Regulations, and any permit conditions that applied.

RIGHT OF REPLY

Before wider circulation this report should be sent to Ambjent Malta and Fondazzjoni Wirt Artna with a fixed deadline for comment. Responses, and any evidence supplied, should be appended and the verdict revisited. If new evidence moves the verdict, this document will be re-issued with a change log (Appendix B).

8 Limitations

- This is a targeted review, not a systematic one. Searches were run in October 2026 and may have missed relevant work.
- We found no studies of short-term, temporary concrete contact with roots. The core of Ambjent’s claim is therefore assessed by mechanism and guidance, not by direct evidence.
- The Times of Malta article was read in full from a supplied PDF. The Newsbook page blocks automated access, so we read only a search excerpt. For several papers we read abstracts or summaries, and some sources are known only through other papers (marked ♦).
- Few studies examine concrete over a membrane around old Mediterranean trees. We extrapolate from pavement and soil-sealing research, mostly from wetter sites with different soils.
- No site visit or tree assessment was done; we saw published photographs only.
- Evidence-map positions are our qualitative judgement. Corrections and additional studies are welcome.

References

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Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the science is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** before wider circulation and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.

Appendix B: revision log

Version	Date	Change
1.0	2 Oct 2026	First issue. Based on the headline and secondary reporting; the full article was not accessed. Treated the claim as a blanket statement that concrete will not harm trees.
1.1	2 Oct 2026	Incorporates the full Times of Malta article. Claim restated as conditional on the concrete being temporary. Added a check of Ambjent's own reasoning, a new contested question on temporary cover, and sub-claims B, C and F. Photo description revised. Added references [17] and [18]. Verdict label unchanged (Not substantiated); the reasoning changed from <i>blanket claim overstated</i> to <i>conditional claim plausible but unevidenced</i> .